Inverted MAML Card 1: Round 4 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.05, L2=1792, Mom=0.85, Clamp=±70 LSB

1. Round 4 Summary & Cumulative Comparison (R1–R4)

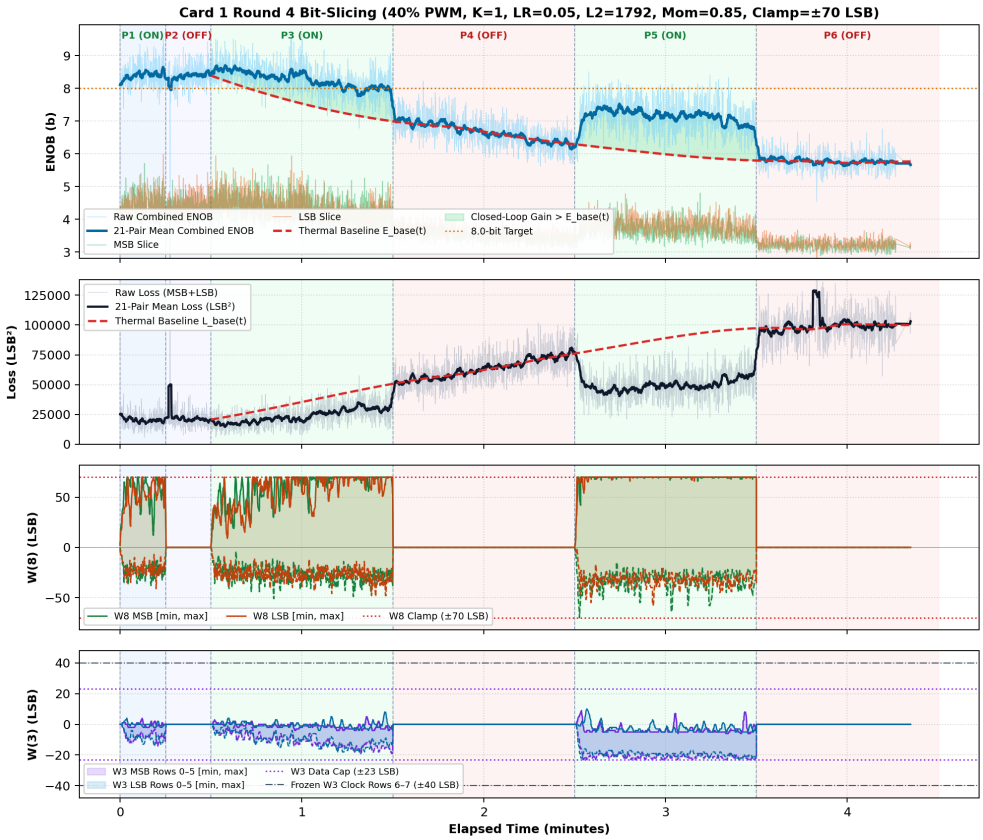
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 4 Detrended Trend Metrics:** Lift vs E_base = **+1.08 b**, Loss Ratio = **1.78x**, Climb = **+0.97 b**, Jitter = **0.40 b**, Recovery = **0.50**, Score = **+2.480**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1dc_fix	0.03	1024	0.9	±75	+1.00b	1.71x	+0.95b	0.40b	0.00	+1.799
R2dc_fix	0.035	1536	0.85	±75	+0.99b	1.69x	+1.16b	0.39b	0.00	+1.957
R3dc_fix	0.02	2560	0.8	±70	+0.04b	1.00x	+0.24b	0.34b	1.00	+0.204
R4dc_fix*	0.05	1792	0.85	±70	+1.08b	1.78x	+0.97b	0.40b	0.50	+2.480

2. Phase-by-Phase Telemetry (3,871 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	279	8.18->8.45 (8.40b)	-	24114->19771	[70,70]	14	279
P2	Cold / Regler OFF	300	8.30->8.42 (8.35b)	-	22607->20779	[0,0]	0	0
P3	Warm / Regler ON	1077	8.51->8.01 (8.28b)	+0.86 b	19556->29278	[70,70]	19	1077
P4	Warm / Regler OFF	895	7.02->6.40 (6.67b)	-0.01 b	52248->71776	[0,0]	0	0
P5	Warm / Regler ON	768	6.38->7.04 (7.15b)	+1.19 b	71157->52465	[70,70]	23	768
P6	Warm / Regler OFF	552	5.79->5.76 (5.77b)	+0.02 b	97639->99490	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 4

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±70 LSB, while W(3) rows 0–5 stay within ±23 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 4 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

